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AI4SmallFarms: A Data Set for Crop Field Delineation in Southeast Asian Smallholder Farms

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***Abstract*—Agricultural field polygons within smallholder farming systems are essential to facilitate the collection of geo-spatial data useful for farmers, managers, and policymakers. However, the limited availability of training labels poses a challenge in developing supervised methods to accurately delineate field boundaries using Earth Observation (EO) data. This letter introduces an open data set for training and benchmarking machine learning methods to delineate agricultural field boundaries in polygon format. The large-scale data set consists of 439,001 field polygons divided into 62 tiles of approximately 5×5 km distributed across Vietnam and Cambodia, covering a range of fields and diverse landscape types. The field polygons have been meticulously digitized from satellite images, following a rigorous multi-step quality control process and topological consistency checks. Multi-temporal composites of Sentinel-2 (S2) images are provided to ensure cloud-free data. We conducted an experimental analysis testing a state-of-the-art Deep Learning (DL) workflow based on fully convolutional networks, contour closing, and polygonization. We anticipate that this large-scale data set will enable researchers to further enhance the delineation of agricultural fields in smallholder farms and to support the achievement of the Sustainable Development Goals (SDG). The data set can be downloaded from https://doi.org/10.17026/dans-xy6-ngg6.**

***Index Terms*—Crop Field Boundaries, Smallholder Farms, Deep Learning (DL), Cambodia, Vietnam, Sentinel-2 (S2) data.**

I. Introduction

Crop field polygons enable digital agriculture services and can record specific information such as crop type, soil characteristics, yield, and farming practices in a spatial database [1]. However, field boundaries are not yet available in many countries, especially in Asian and African regions where smallholder farms with fields smaller than two hectares comprise 70% of the cropland [2]. This information gap hampers the achievement of Target 2.3 of the United Nations Sustainable Development Goals (SDGs), which emphasise the need to improve the agricultural productivity of small-scale food producers to achieve food security, improved nutrition and sustainable agriculture. Moreover, crop field boundaries often correspond to visible cadastral parcels, which are essential parts of the country's land administration systems. Land tenure security offers many benefits to farmers, such as access to credit and investments, government and insurance services, and reduced conflicts. The importance of secure tenure rights to land, with legally recognized documentation, is recognized by Target 1.4 of the SDG agenda as an essential factor in ending poverty.

Recent developments in Deep Learning (DL) and Earth observation show that field boundaries can be effectively delineated using Very High resolution (VHR) images [1], [3]. However, the main bottleneck that limits the generalization ability and general performance of such methods in smallholder farms is the lack of a large amount of reference data [4]. While several benchmark data sets focus on crop type mapping in Europe [5], [6], [7] and a few in Africa [8], [9], little has been done so far for crop boundary delineation. In [10], the authors introduced a data set of images and field boundary labels collected from seven European countries representing 14.8M parcels and covering 372K km^2. It is far more challenging to access crop field boundary data in smallholder farms that are common in many low- and middle-income countries. The authors in [4], publicly released a data set containing 10,000 Indian field boundary labels. However, only a fraction of the fields in each image is labeled.

In this letter, we introduce *AI4SmallFarms*, a benchmark data set for the automated extraction of field polygons in small-scale farms located in two South Asian countries: Vietnam and Cambodia. These countries are characterized by fragmented agricultural areas with small fields of less than one hectare. Although VHR images can provide accurate results, their high acquisition cost hinders the use of commercial images for extracting and updating crop boundaries. Moreover, their restricted policies prevent the creation of open data sets. Therefore, we explore here the potentials of Sentinel-2 (S2) data for field boundary delineation [11], which are available openly and freely to all users. Nevertheless, the coarser resolution of 10m poses significant challenges in accurately delineating the small-size fields present in smallholder farms.

To the best of our knowledge, *AI4SmallFarms* is the first large-scale open data set available to the public for smallholder farming in Southeast Asia. The data set aims to support the development of operational mapping and monitoring systems of crop boundaries in Asia. It also aims to facilitate the development of machine learning methods and support the achievement of SDG 2 "Zero Hunger" and ongoing initiatives of the Food and Agriculture Organization (FAO) fostering the implementation of effective food security measures. This activity has been carried out in collaboration with the Image Analysis and Data Fusion (IADF) Technical Committee of the IEEE Geoscience and Remote Sensing Society (GRSS).

II. Study Areas and Benchmark Data Set

Although agriculture is a key sector in Vietnam and Cambodia, involving a large portion of their population, these countries lack comprehensive, high-quality, and up-to-date field boundary data. Having rice paddy occupying nearly 80% of the harvested area in Cambodia, and Vietnam being one of the largest rice exporters at the global level, both countries are

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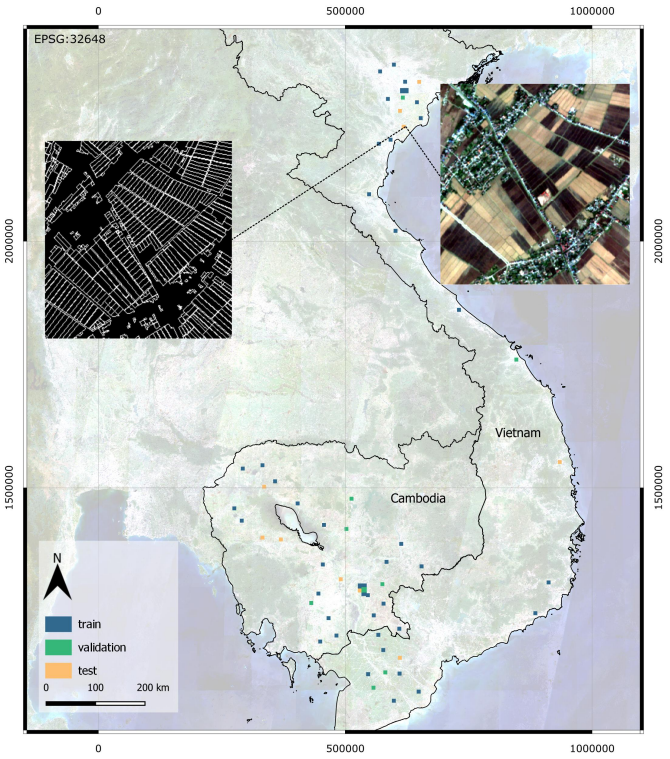


Fig. 1: Spatial distribution of the training, validation, and test tiles shown in blue, yellow, and green, respectively. For one tile, the S2 image and the corresponding reference boundaries are displayed.

mainly focused on rice production. Their agricultural system is predominantly characterized by small-scale farming, with subsistence farming systems located in rural areas.

A. Data Set Organization

The proposed *AI4SmallFarms* data set is made up of 62 tiles having a size of approximately 5×5 km, where professional photo interpreters manually digitized all the visible agricultural fields. Figure 1 shows the spatial distribution of the reference tiles across most of the agricultural regions present in Vietnam and Cambodia. The tiles are split into spatially disjoint subsets for training, validating, and testing machine learning algorithms. Figure 1 also shows an example of a S2 image tile and the corresponding digitized crop boundaries.

Table I shows the number of tiles, the number of field polygons, and the average crop size present in the training, validation, and test sets. The reference boundary are manually digitized resulting in 318,088 and 120,913 field polygons in Cambodia and Vietnam, respectively. Differently from [4], all the visible crop boundaries within each tile are manually digitized, resulting in a total number of 439,001 polygons.

The field boundaries are published in a vector format as polygons and polylines. Using the geo-referenced vector data, any EO satellite data can be associated with the crop field boundaries. Details of the delineation procedure are provided in Section IIB. The benchmark data set includes also preprocessed S2 imagery (details in subsection IIC).

TABLE I: Number of tiles, field polygons and average crop size in each split of the proposed benchmark dataset.

	N. of Tiles	N. of Polygons	Avg. Crop Size (m^2)
Training	43	304777	2920
Validation	9	83066	2385
Test	10	51158	4042
Total	62	439001	2950

To facilitate the training process on S2 data, a freely available agricultural field boundary data set (Basisregistratie Gewaspercelen (BRP)) and S2 data from the Netherlands are used to pre-train the DL-based workflow. This data set consists of 87 tiles of 10×10 km^2 and 192,321 field polygons and is released together with *AI4SmallFarms* to enable the reproducibility of the experiments.

To assess the impact of the spatial resolution in delineating crop boundaries in the considered complex agricultural area, experiments are carried out also considering freely downloadable VHR Google Map (GM) RGB images with a spatial resolution of about 0.5m (not included in the benchmark data set because of redistribution policy restrictions). The satellite imagery along with the reference polygons are used to create image/label pairs. This is done for all the plots in Vietnam and Cambodia and for both S2 data and VHR GM data [1]. For the purpose of training and accuracy assessment, the rasterized reference boundaries of the VHR GM data are buffered with a distance of 1m, while S2 reference data are not buffered.

B. Digitization Method

The digitization of reference polygons has been carried out by visual interpretation of S2 and VHR GM images acquired in August 2021[2]. Each tile is fully digitized, including all visible agricultural boundaries. The process followed a rigorous multi-step approach involving two quality control gates and topological consistency checks. First, agricultural field boundaries are visually detected and digitized in polygon format by the production team. Second, all polygons are checked by an independent quality control team to verify shape correctness, missing polygons, and software-driven checks to identify topological errors (e.g., to avoid overlap between polygons or unnecessary gaps between adjacent fields). All minor errors have been corrected by the quality control team. In case of major errors, data has been reverted to the production team for re-work. Third, 10% of the vectors have been further checked by a second quality control team to verify the projection system, data completeness, nomenclatures of files, and output format.

C. S2 Satellite Data

AI4SmallFarms is designed to advance the capability of using S2 data for crop boundary delineation in smallholder

[1]Processed in FME Workbench 2023.0

[2]The digitization process has been carried out with the support of RMSI https://www.rmsi.com/ in collaboration with IADF and with the financial support of GRSS.

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farms. To ensure the reproducibility of our experimental analysis and foster further developments, the data set includes the preprocessed S2 images used to perform the experiments reported in this letter. Because of the heavy cloud coverage typically present in the considered study areas, the atmospherically corrected S2 L2A images are preprocessed to generate monthly composites almost free of clouds using the standard statistics-based approach described in [6]. This approach aggregates all the S2 images acquired within a month into a composite by calculating the median pixel value, excluding the pixels that are affected by clouds or shadows. In the considered experimental analysis, we focused on the least cloudy months of the dry season, i.e., mainly January and February (and also November, December and May in some cases), by selecting only the S2 images having cloud cover $< 40\%$. For each tile, the monthly composites obtained were visually inspected to select the least cloudy one. Only the blue (B2–490 nm), green (B3–560 nm), red (B4–665 nm), and Near Infrared (B8-842 nm) bands are considered, since these are the most relevant spectral bands for crop boundary delineation and the ones acquired at the highest spatial resolution, i.e., 10m.

III. Experimental Analysis

We conducted an experimental analysis using both the S2 and GM images with a state-of-the-art DL-based workflow inspired from [1] and depicted in Figure 2 consisting of three main consecutive steps: 1) semantic segmentation using a Fully Convolutional Network (FCN), 2) contour closing, and 3) polygonization.

The first step generates pixel-wise predictions of boundary probability scores. For this purpose, we adopted a UNet-like architecture [12] using the implementation available at[3]. Since input image/label tiles have varying sizes, the data were split into patches of fixed dimensions. We used patches of 256×256 pixels for S2, and 512×512 for the GM images. We compared three models: 1) trained from scratch with S2 data, 2) pre-trained in the Netherlands with S2 data and fine-tuned on *AI4SmallFarms*, and 3) trained from scratch with GM data. The hyper-parameters of the three networks were tuned using the validation set. All models were trained with the Adam optimizer (initial learning rate = 0.001, decay step = 10000, decay rate = 0.9), a focal cross-entropy loss function, and an early stopping procedure (max 1000 epochs). Table II reports the values of the other hyper-parameters.

The output of the FCN does not guarantee obtaining closed contours and often results in fragmented lines that cannot be directly converted to polygons. For this reason, we adopted a contour closing procedure that allows us to derive a closed segment for each field [4]. The procedure consists of the following steps: 1) extended-minima transform, 2) impose minima, 3) connected component labeling, and 4) watershed transform. In the polygonization step, the closed contours were converted to vector format (polylines), simplified with the Douglas-Peucker algorithm and finally, converted to polygons[5].

[3]https://deepsense.ai/deep-learning-for-satellite-imagery-via-image-segmentation

[4]https://imagej.net/plugins/morphological-segmentation

[5]Processed in ArcGIS Pro 3.1.1

TABLE II: Selected hyper-parameters of the three models.

	S2 (scratch)	**S2 (pretrained)**	**GM**
Batch size	8	2	8
Segmentation tolerance	0.05	0.1	0.1
Simplify tolerance	10	10	2

TABLE III: Quantitative average results across all test tiles.

	Precision	**Recall**	**F1-score**	**PoLiS (m)**
S2 (from scratch)	0.42	0.36	0.38	27.0
S2 (pre-trained)	0.48	0.33	0.39	26.7
GM Images	0.47	0.52	0.49	20.3

To assess the accuracy of the predicted boundaries, both raster-based and vector-based metrics are used. The confusion matrix is used to calculate raster metrics, i.e., precision, recall, and F1-score of the class boundary. These metrics are calculated per tile using the binary output of the contour closing step and the corresponding reference boundaries. However, raster metrics do not fully capture the agreement between polygons. Therefore, we propose to adopt the PoLiS metric [13], which analyses the difference in position and shape between predicted and reference polygons. Since the computation of PoLiS takes matched pairs of the prediction-reference polygons as input, its value is independent of the spatial resolution and the thickness of the raster boundaries, which critically affects the raster-based metrics. For this reason, we consider PoLiS a more suitable metric for field boundary delineation tasks, especially when comparing results obtained from images of different resolutions.

IV. Experiment results

Quantitative results: Table III shows the quantitative results of our experiments using S2 (with and without pre-training) and GM images as input to our workflow. For the S2 experiments, we observe that the model pre-trained on the Dutch data set and fine-tuned in Cambodia and Vietnam resulted in an average F1-score of 0.39 and an average PoLiS distance of 26.7m, and is only slightly superior to the model trained from scratch, which results in an average F1-score of 0.38 and an average PoLiS distance of 27.0m. This result shows that our benchmark data set is large enough to obtain accurate results without the need for pretraining.

The average F1-score of the GM experiment is 0.49. The high spatial resolution of the GM results in an average PoLiS value of 20.3m, which is significantly better than the average PoLiS value of 26.7m obtained using S2 data.

Qualitative results: We selected two tiles (tile 12 and 27) to illustrate and assess the quality of the predicted polygons. Figure 3 shows the reference data, the FCN output (boundary score), and the final vector predictions for both tiles. As expected, vector predictions from the GM experiment are more regular compared to the predictions from the S2 experiment. Next to that, the boundary scores from the S2 experiment are sometimes too coarse, resulting in missing boundaries. These differences are mainly due to the higher spatial resolution of the GM data. Both differences can also be observed in

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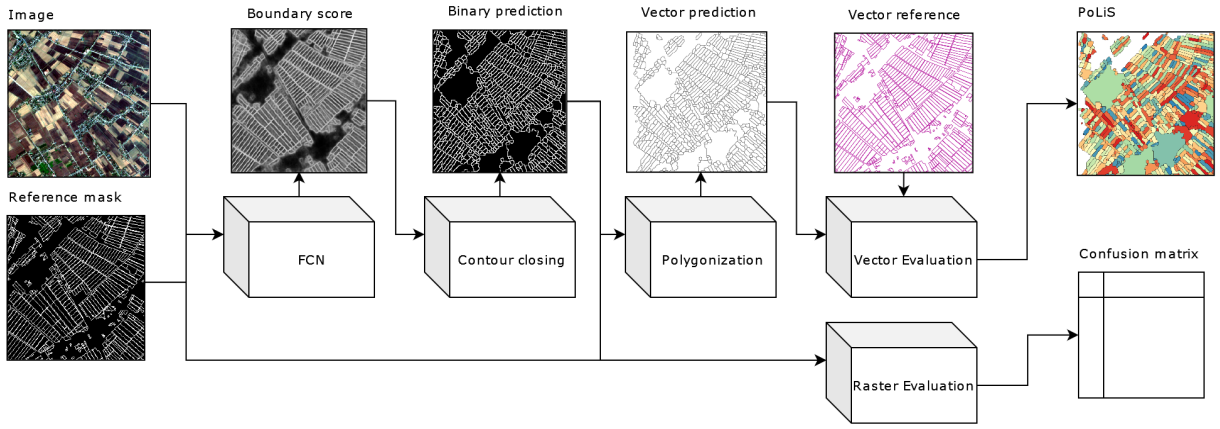


Fig. 2: Overview of the adopted DL-based workflow for field boundary delineation in polygon format.

the calculated PoLiS metric. The average PoLiS distance for tile 12 is 11.1m and 17.1m for the GM and S2 predictions, respectively. For tile 27, the average PoLiS metric is 19.4m and 35.0m for the GM and S2 predictions, respectively. Figure 4 shows a closeup of the results on tile 12. The GM predictions almost fully cover the reference data and are relatively regular, whereas the S2 predictions have a lot of missing boundaries and are not as regular as the GM predictions.

V. Challenges

This section summarizes the main challenges posed by the proposed data set:

Crop Field Size: Due to the small average crop sizes (far below one hectare), it is difficult to accurately delineate crop boundaries in smallholder farms using S2 images. However, these free and open-access data enable the regular update of crop boundary databases. For this reason, we see the need for developing super-resolution semantic boundary detection strategies to further increase the spatial resolution of results, e.g., by exploiting the spatial-contextual information in label space [11] or adopting generative models.

Satellite Data Preparation: Because of the tropical climate, these areas are typically affected by heavy cloud cover. This may hamper the possibility of using specific temporal acquisitions that can emphasize the phenological state of the crops. The least cloudy period is the dry season, when crops may exhibit low contrast to the background, leading to poor crop delineation results. Possible future work may investigate the use of Synthetic Aperture Radar (SAR) data.

Imbalanced Classification Problem: The presented crop boundary delineation task poses an extremely inbalanced classification problem due to the prevalence of non-boundary pixels. For this reason, we adopted the focal cross-entropy loss in our experiments. Nevertheless, developing a loss function tailored to the specific properties of the crop boundary delineation task is expected to improve the result and mitigate the extreme class imbalance.

Crop Variability: *AI4SmallFarms* consists of tiles showing substantial variability among fields, different landscape conditions, soil characteristics, crop management practices and crop arrangements. This data set provides an opportunity to assess the generalization capabilities of crop boundary delineation approaches. Indeed, it is hard for a single model to achieve accurate delineation results across all the considered tiles. This data set aims to support the development of robust models that show improved generalization ability when encountering unseen landscapes. Future studies may explore advanced deep learning solutions based on semi-supervised learning or self-training. Moreover, we expect that recent semantic segmentation models based for example on vision transformers (e.g., Swin Transformer) may further improve the feature extraction performance.

VI. Conclusion

This paper presented *AI4SmallFarms*, a large-scale data set for field boundary delineation in fragmented agricultural areas characterized by small fields. The data set comprises 439,001 field polygons divided into 62 non-overlapping tiles of approximately 5×5 km distributed across Vietnam and Cambodia. The field polygons were meticulously digitized using S2 and GM images, following a rigorous multi-step quality control procedure and topological consistency checks. For each tile, all visible crops were manually digitized. To ensure the reproducibility of the experimental results, the data set is provided into three spatially disjoint sets, i.e., training, validation, and test tiles. The experimental results obtained with a state-of-the-art DL-based workflow highlights the main challenges of the proposed data set and its properties. *AI4SmallFarms* is publicly available to support the community in advancing the development of supervised machine learning methods for crop boundary delineation in agricultural areas with smallholder farms using S2 data.

VII. Acknowledgements

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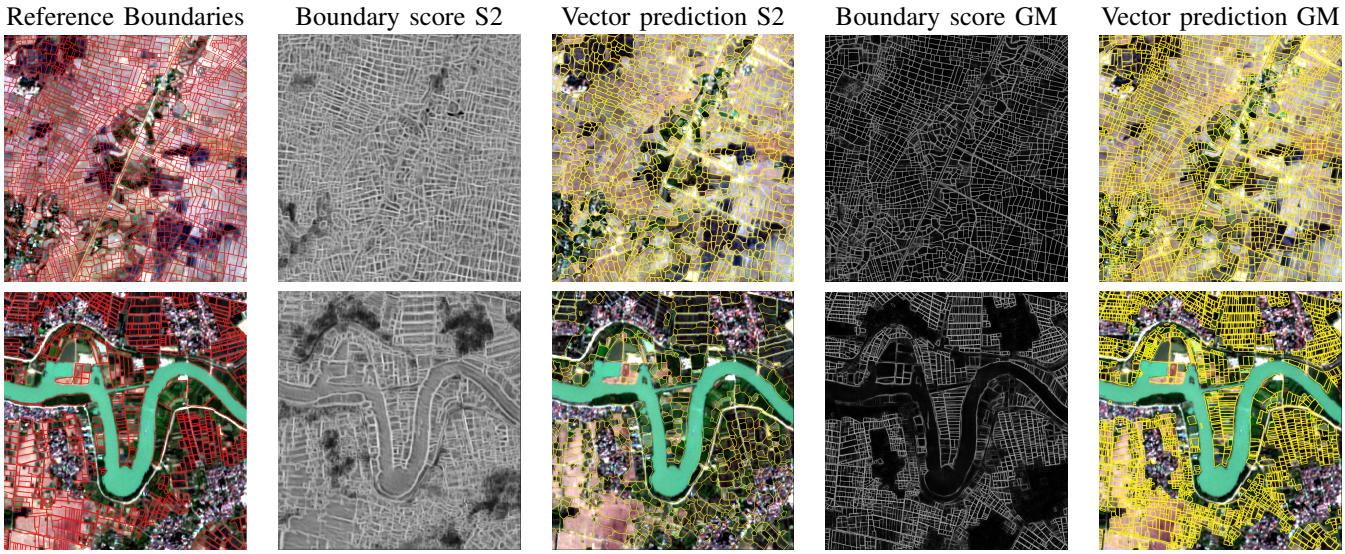
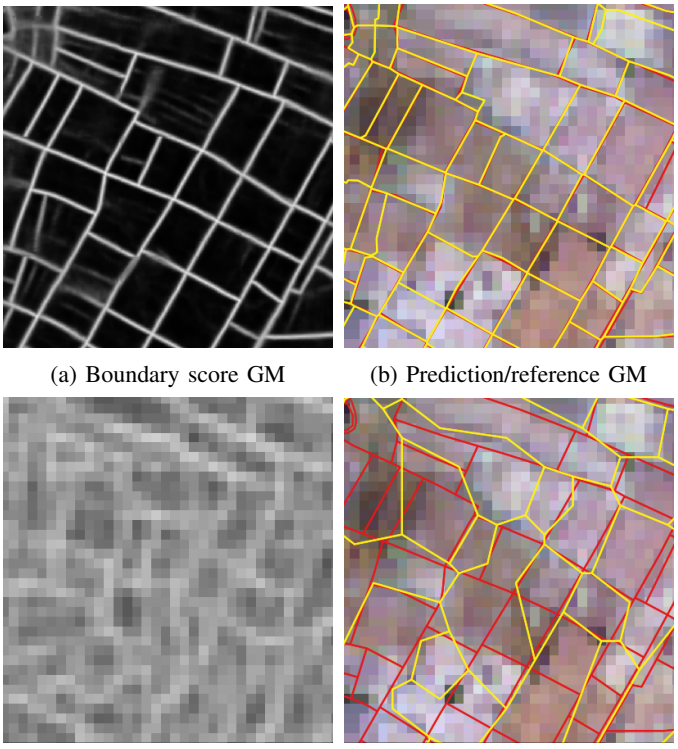


Fig. 3: Qualitative results of tiles Cambodia 12 (top) and Vietnam 27 (bottom).



(a) Boundary score GM (b) Prediction/reference GM

(c) Boundary score S2 (d) Prediction/reference S2

Fig. 4: Closeup of tile Cambodia 12 showing the limitation of S2 with respect to GM in capturing small-sized fields. Reference boundaries are in red and predictions in yellow.

and the creation of *AI4SmallFarms* benchmark. Thanks to the company RMSI for the meticulous work.

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